1(a) 4 properties of distribution f^ 1,1

1) $0 \leq F_X(x) \leq 1 \quad \forall x \in \mathbb{R}$

2) $\lim_{x \to -\infty} F_X(x) = 0$ & $\lim_{x \to \infty} F_X(x) = 1$ (2)

3) Non-decreasing

if $x \geq y \Rightarrow F_X(x) \geq F_X(y)$

4) Right continuous, Left Limits

b) To be non-decreasing we need

$$F_X(1) \geq \frac{1}{3}$$

So need $1 - c \times \frac{1}{1} \geq \frac{1}{3} \Rightarrow c \leq 2/3$ (1)

To be bound by 1, ie $F_X(x) \leq 1 \; \forall x$

we need $c \geq 0$ (1)

So $0 \leq c \leq \frac{2}{3}$

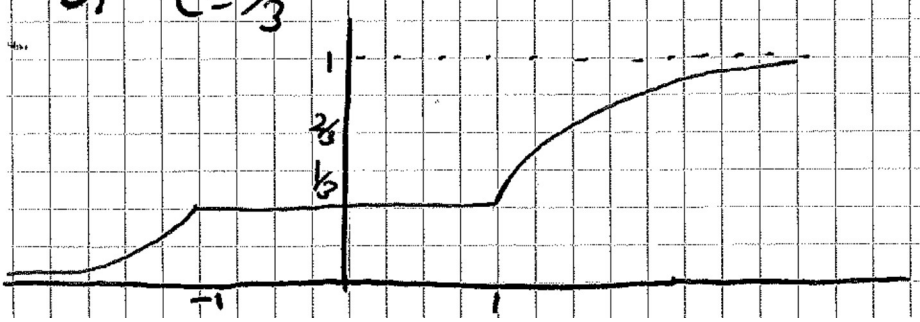
This then gives a f^ which is non-negative

i.e. $1 - \frac{c}{x^2} \geq \frac{1}{3} \geq 0$ (1)

and is right continuous with left limits

and $\frac{c}{x^2} \to 0$ as $x \to \infty$ so $1 - \frac{c}{x^2} \to 1$ as $x \to \infty$

c) $c = 2/3$



(1)

c) If $c = 2/3$, F_X is a continuous fn & differentiable almost everywhere so X has a density $f_X(x) = F_X'(x)$ ①

$$f_X(x) = \begin{cases} -\frac{1}{3x^{\#3}} \times 4 \\ 0 \\ +\frac{2}{3x^{\#3}} \times 2 \end{cases} = \begin{cases} -\frac{4}{3x^{\#5}} & x \leq -1 \\ 0 & -1 < x < 1 \\ \frac{4}{3x^{\#3}} & x \geq 1 \end{cases}$$ ①

d) The image of X is when $f_X(x) > 0$ ~~from~~

or image of X is when $F_X(x)$ is increasing

from either the definition of $F_X(x)$ or from part (b) ①

this is when $\text{Im}X = (-\infty, -1] \cup [1, \infty)$

e

e) For continuous r.v the expected value is

$$E(X) = \int_{\mathbb{R}} x f_X(x)\, dx \quad ①$$

So here $E(X) = \int_{-\infty}^{-1} x \times -\frac{4}{3x^5}\, dx + \int_1^{\infty} x \times \frac{4}{3x^3}\, dx$ ①

$$= \int_{-\infty}^{-1} -\frac{4}{3x^4}\, dx + \int_1^{\infty} \frac{4}{3x^2}\, dx$$

$$= \left[-\frac{4}{3x^3} \times -\frac{1}{3}\right]_{-\infty}^{-1} + \left[\frac{4}{3x} \times -1\right]_1^{\infty} = \frac{4}{3} - \frac{4}{9} = \frac{8}{9}$$ ①

5 (i) We have $\text{Im}X = (-\infty, -1] \cup [1, \infty)$

Let $Y = X^2$

Under squaring we have $(-\infty, -1]$ maps to $[1, \infty)$
and $[1, \infty)$ maps to $[1, \infty)$

So the image of Y is $[1, \infty)$ ①

Now distribution f^n of Y

$$F_Y(y) = P(Y \leq y) \quad \text{by definition}$$
$$= P(X^2 \leq y) \quad \text{as } Y = X^2 \ ①$$
$$= P(-\sqrt{y} \leq X \leq \sqrt{y}) \quad \text{as rearranging inequality to make } X \text{ subject } ①$$
$$= P(X \leq \sqrt{y}) - P(X \leq -\sqrt{y}) \quad \text{(note as } X \text{ is continuous } P(X < \sqrt{y}) \text{ is the same as } P(X \leq -\sqrt{y}))$$
$$= F_X(\sqrt{y}) - F_X(-\sqrt{y}) \quad \text{by definition of } F_X \ ①$$

(ii) We have Y is continuous so

$$f_Y(y) = F'_Y(y) = \frac{d}{dy}\left(F_X(\sqrt{y}) - F_X(-\sqrt{y})\right) \ ①$$

$$= f_X(\sqrt{y}) \times \frac{1}{\sqrt{y}} \times \frac{1}{2} + f_X(-\sqrt{y}) \times \frac{1}{\sqrt{y}} \times \frac{1}{2}$$

①

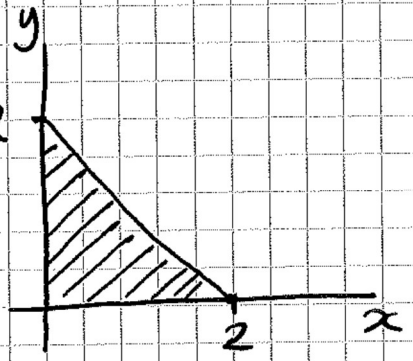
ii) So $f_Y(y) = \frac{1}{2\sqrt{y}} \left(f_X(\sqrt{y}) + f_X(-\sqrt{y}) \right)$

$$= \frac{1}{2} y^{-\frac{1}{2}} \times \left(\frac{4}{3} \times \frac{1}{y^{3/2}} + \frac{-4}{3} \times \frac{1}{-y^{5/2}} \right)$$

$$= \frac{2}{3} \left(\frac{1}{y^2} + \frac{1}{y^3} \right) \text{(2)}$$

2)

a)



①

b) $f_X(x) = \int f_{X,Y}(x,y)\,dy$, def^n of marginal dist ①

So here $f_X(x) = \int_0^{2-x} \frac{3}{4}y\,dy = \left[\frac{3}{4}y^2 \times \frac{1}{2}\right]_0^{2-x}$

$= \frac{3}{8}(2-x)^2$ for $x \in [0,2]$ ①

The conditional density is given by

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} \quad ①$$

So $f_{Y|X}(y|x) = \frac{\frac{3}{4}y}{\frac{3}{8}(2-x)^2} = \frac{2y}{(2-x)^2}$ for $0 \le x,y$ & $0 \le x+y \le 2$ ①

i) $P(Y<1 \mid X=\frac{1}{2}) = \int_0^1 f_{Y|X}(y|\tfrac{1}{2})\,dy$ ①

$= \int_0^1 \frac{2y}{(\frac{3}{2})^2}\,dy = \frac{8}{9}\left[\frac{y^2}{2}\right]_0^1 = \frac{4}{9}$ ①

ii) $E(Y|X=\frac{1}{2}) = \int_0^{3/2} y\, f_{Y|X}(y|\tfrac{1}{2})\, dy$ ①

$= \int_0^{3/2} \frac{2y^2}{(\frac{3}{2})^2}\, dy = \left[\frac{4}{9} \times \frac{2}{3} \times \left(\frac{3}{2}\right)^3\right]$

$= 1$ ①

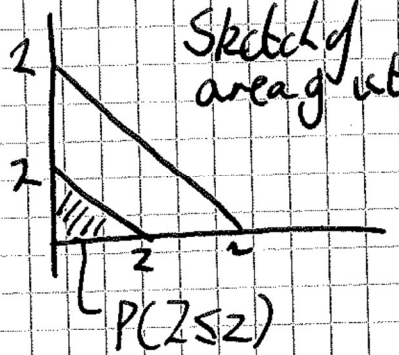
c) Let $Z = X + Y$

We have $\text{Im } Z = [0, 2]$ ①

Now we want to find distribution f^n of Z

$F_Z(z) = P(Z \leq z)$ ①

$= \int_0^z \int_0^{z-y} f_{XY}(x, y)\, dx\, dy$



$= \int_0^z \int_0^{z-y} \frac{3}{4} y\, dx\, dy$ ①

$= \int_0^z \frac{3}{4} y(z-y)\, dy = \int_0^z \frac{3}{4}(zy - y^2)\, dy$

$= \int_0^z \frac{3}{4}\left(\frac{zy^2}{2} - \frac{y^3}{3}\right)\Big]_0^z$ ①

$= \frac{3}{4}\left(\frac{z^3}{2} - \frac{z^3}{3}\right) = \frac{z^3}{8}$

So $F_Z(z) = \begin{cases} 0 & \text{if } z < 0 \\ \frac{z^3}{8} & \text{if } 0 \le z < 2 \\ 1 & \text{if } 1 < z \end{cases}$ ①

Now this is continuous and differentiable almost everywhere

So the density of Z,

$$f_Z(z) = \frac{d F_Z(z)}{dz} = \begin{cases} \frac{3z^2}{8} & 0 \le z \le 2 \\ 0 & \text{otherwise} \end{cases}$$

① ①

3a i) We have $X = 15 + Y$ and $Y \sim Exp(\frac{1}{10})$

So Bell's arrival after 9am given he leaves at 8.45am will have the same distribution as Y.

The meeting starts when both have arrived, i.e when the 2nd person arrives. So this is the maximum of their arrival time after 9am, $\max(W, Y)$ (1)

So $T = \max(Y, W)$

We have $Im\, Y = [0, \infty)$ & $Im\, W = [0, 15]$

So $Im\, T = [0, \infty)$ (1)

ii) We have distribution f^n of T

$$F_T(t) = P(T \leq t)$$
$$= P(\max(Y, W) \leq t)$$
$$= P(Y \leq t \ \& \ W \leq t) \quad \text{as if } \max(Y,W) \leq t \text{ then } Y \leq t \ \& \ W \leq t$$
$$= P(Y \leq t)\, P(W \leq t) \quad \text{by independence (1)}$$
$$= F_Y(t)\, F_W(t)$$

$Y \sim Exp(\frac{1}{10})$ so $F_Y(y) = \begin{cases} 0 & y < 0 \\ 1 - e^{-y/10} & 0 \leq y \end{cases}$ (1)

$W \sim U[0, 15]$ so $F_W(w) = \begin{cases} 0 & w < 0 \\ w/15 & 0 \leq w < 15 \\ 1 & 15 \leq w \end{cases}$ (1)

So $F_T(t) = \begin{cases} 0 & t<0 \\ (1-e^{-t/10})(\frac{t}{15}) & 0 \leq t < 15 \\ (1-e^{-t/10}) & 15 \leq t \end{cases}$ ①

iii) $P(T<15 \mid T<30)$

$= \frac{P(T<15 \ \& \ T<30)}{P(T<30)}$ ① conditional probability def^n

$= \frac{P(T<15)}{P(T<30)}$ ① as $T<15 \Rightarrow T<30$

$= \frac{F_T(15)}{F_T(30)} = \frac{1-e^{-3/2}}{1-e^{-3}} = 0.818$ ①

b) We need to find x s.t

$P(X>x) = 0.01$ ①

By def^n we have

$P(Y+15>x) = 0.01$

Since $Y \sim Exp(\frac{1}{10})$ we also solve ①

$e^{-\left(\frac{x-15}{10}\right)} = 0.01$

So $x = 61.05$

Therefore she needs to leave at 8.58. ①

4) $X \sim \text{Gamma}(\alpha, \lambda)$ $\quad Im X = [0, \infty)$

So $f_X(x) = \begin{cases} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} & 0 \le x \\ 0 & \text{otherwise} \end{cases}$

So M.G.F for X

$$M_X(\theta) = \int_0^\infty e^{\theta x} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \, dx = E(e^{\theta X}) \quad (1)$$

$$= \int_0^\infty \lambda^\alpha \frac{x^{\alpha-1} e^{(\theta-\lambda)x}}{\Gamma(\alpha)} \, dx \quad (1)$$

$$= \frac{\lambda^\alpha}{(\lambda-\theta)^\alpha} \int_0^\infty \underbrace{\frac{(\lambda-\theta)^\alpha x^{\alpha-1} e^{-(\lambda-\theta)x}}{\Gamma(\alpha)}} \, dx$$

density for $\text{Gamma}(\alpha, \lambda-\theta)$ r.v so integrates to 1 (1)

$$= \left(\frac{\lambda}{\lambda-\theta}\right)^\alpha \quad (1)$$

b) Let $Y = cX$

We have M.G.F for Y

$$M_Y(\theta) = E(e^{\theta Y}) \quad \text{def}^n \text{ of } M_Y(\theta)$$

$$= E(e^{\theta c X}) \quad \text{as } Y = cX \quad (2)$$

$$= E(e^{(\theta c)X}) = M_X(\theta c) \quad \text{def of } M_X(\theta)$$

Now from a) we have $M_X(\theta) = \left(\frac{\lambda}{\lambda-\theta}\right)^{\alpha}$

So $M_Y(\theta) = M_X(c\theta) = \left(\frac{\lambda}{\lambda - c\theta}\right)^{\alpha}$ ①

$= \left(\frac{\lambda/c}{\lambda/c - \theta}\right)^{\alpha}$ multiply top & bottom by $1/c$

But this is the moment generating fn for Gamma$(\alpha, \lambda/c)$ ①

Since M.G.F uniquely define the distribution of Y

Y has a Gamma$(\alpha, \frac{\lambda}{c})$ distribution. ①

5a) $N \sim Po(150)$

$X_i \sim Bernoulli(0.3)$

By def^n $N_N = \sum_{i=1}^{N} X_i$ a compound r.v.

Since compound sum we have

$$M_{N_N}(\theta) = G_N(M_X(\theta)) \quad (1)$$

where $M_{N_N}(\theta)$ is m.g.f of N_N

$M_X(\theta)$ is m.g.f of X

$G_N(\theta)$ is p.g.f of N

We have $N \sim$ Poisson (150)

so $G_N(\theta) = E(\theta^N) = e^{-150(1-\theta)}$ (1)

$X \sim Bernoulli(0.3)$

So $M_X(\theta) = E(e^{\theta X}) = e^{\theta}_{x} P(X=1) + e^{\theta \cdot 0} P(X=0)$

$= 0.7e^{\theta} + 0.3$ (1)

So $M_{N_H}(\theta) = e^{-150(1-(0.3e^{\theta}+0.7))}$

$M_{N_H}(\theta) = e^{-45(1-e^{\theta})} = e^{45(e^{\theta}-1)}$ (1)

But $M_{N_H}(\theta)$ is the m.g.f for a Poisson r.v. with $\lambda = 45$

so by consequence $N_H \sim Po(45)$ (1)

b) We have $F_{S|N}(s|n) = P(S \leq s \mid N=n)$

Note $Im\, S|N = [0,1]$ for all n so $Im\, S = [0,1]$

We want to find

$$F_S(s) = P(S \leq s)$$

We use the partition theorem (1)

So $P(S \leq s) = \sum_{n=0}^{\infty} P(S \leq s \mid N=n) \times P(N=n)$ (1)

$= \sum_{n=0}^{\infty} s^{n/10} \times P(N=n)$ from question

$= \sum_{n=0}^{\infty} s^{n/10} \times \frac{e^{-45}45^n}{n!}$ (1) p.m.f for $Po\,45$

$= \sum_{n=0}^{\infty} \left(s^{\frac{1}{10}}\right)^n \frac{e^{-45}45^n}{n!}$

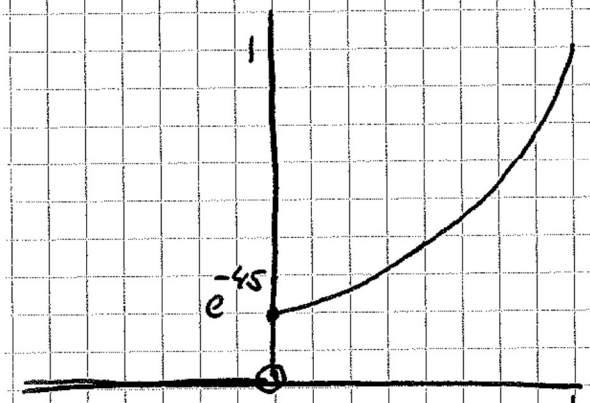
$= \sum_{n=0}^{\infty} e^{-45} \frac{(45s^{\frac{1}{10}})^n}{n!} = e^{-45} \times e^{45s^{\frac{1}{10}}}$ (1)

↑ exponential sum

So $P(S \leq s) = e^{45(s^{\frac{1}{10}}-1)}$

So the distribution of S is

$$F_S(s) = \begin{cases} 0 & s < 0 \\ e^{45(s^{1/10}-1)} & 0 \le s < 1 \\ 1 & 1 < s \end{cases}$$



(1)

6 a) Since X & Y are independent the joint density is given by

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) \quad (1)$$

So $E(XY) = \iint xy\, f_{XY}(x,y)\, dx\, dy$ def^n

$= \iint xy\, f_X(x) f_Y(y)\, dx\, dy$ by independence (1)

$= \int y f_Y(y) \int x f_X(x)\, dx\, dy$ rearranging integral

$= \int y f_Y(y) E(X)\, dy$ def^n of $E(X)$

$= E(X)E(Y)$ (1)

b) From a) $E(X_1X_2) = E(X_1)E(X_2)$

So need to find

$$E(X_i) = \int_{\frac{1}{e}}^{e} x \times \frac{1}{2x}\, dx = \left[\frac{1}{2}x\right]_{\frac{1}{e}}^{e} = \frac{1}{2}(e - e^{-1}) \quad (2)$$

$$= 1.175$$

So $E(X_1X_2) = 1.38$ (1)

c) Let $Y_i = \ln(X_i)$ $\quad \text{Im } Y_i = [\ln(\frac{1}{e}), \ln(e)]$

$= [-1, 1]$ (1)

So dist of Y_i

$F_{Y_i}(y) = P(Y_i \leq y)$ by def^n

$= P(\ln(X_i) \leq y)$ (1) as $Y = \ln(X_i)$

$= P(X_i \leq e^y)$ solving inequality for X_i

$F_{Y_i}(y) = F_{X_i}(e^y)$ (1)

Now to find density of Y_i

$$f_{Y_i}(y) = \frac{d}{dy}(F_{Y_i}(y)) = \frac{d}{dy}(F_{X_i}(e^y))$$

$$= f_{X_i}(e^y) \times e^y \quad \text{for } y \in [-1,1] \quad (1)$$

$$\text{So } f_{Y_i}(y) = \frac{1}{2e^y} \times e^y = \frac{1}{2} \quad \text{for } y \in [-1,1]$$

$$\text{So } f_{Y_i}(y) = \begin{cases} \frac{1}{2} & -1 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(1)

This is the density function for an exp uniform r.v. on $[-1,1]$. □

d) Central limit theorem

Let $Z_1, Z_2, \ldots$ be i.i.d. r.v. with $E(Z_i) = \mu$ and $\text{Var}(Z_i) = \sigma^2 < \infty$ and let S_n

d) Central Limit theorem

Let $W_1, W_2, \ldots$ be i.i.d. r.v with $E(W_i) = \mu$ and $Var(W_i) = \sigma^2$ with $\sigma^2 < \infty$ and let $S_n = W_1 + \cdots + W_n$ for $n = 1, 2, \ldots$

Then for any $a < b$ (1)

$$\lim_{n\to\infty} P\left(a \leq \frac{S_n - n\mu}{\sqrt{n\sigma^2}} \leq b\right) = P(a \leq Z \leq b)$$

where $Z \sim N(0,1)$

In this case we are interested in

$$P\left(\prod_{i=1}^{200} X_i < 0.083\right)$$

$$= P\left(\sum_{i=1}^{200} \ln X_i < -2.49\right)$$ (1) taking ln of both sides

From part c we have $\ln X_i = Y_i \sim U[0,1]$

so $E(Y_i) = 0$ & $Var(Y_i) = \frac{1}{6}$ (1)

So $P\left(\prod_{i=1}^{200} X_i < 0.083\right)$

$$= P\left(\frac{\sum_{i=1}^{200} Y_i}{\sqrt{\frac{200}{6}}} < \frac{-2.49}{\sqrt{\frac{200}{6}}}\right)$$ (1)

But the CLT tells us $\sum_{i=1}^{200} Y_i / \sqrt{\frac{200}{6}}$ is approx $Z \sim N(0,1)$

So $P\left(\prod_{i=1}^{200} X_i < 0.083\right) \approx P(Z < -0.431) = 0.334$ (1)

from tables